



Luvata hollow conductors – enabling more efficient electric vehicles with direct stator cooling

I. Summary

In order to meet the growing need to decarbonize transportation, the use of electric vehicles (EVs) will need to increase. Direct-cooled motors with hollow conductor stators offer several advantages over the indirect-cooled motors currently used in EVs. More effective cooling of the motor windings enables a higher power output, increased efficiency – especially at partial loads, and the use of lower temperature grade permanent magnets, meaning smaller, more efficient motors can be built using cost-effective off-the-shelf materials.

II. Market background

According to the European Environment Agency, in 2019 transport accounted for nearly 30% of the EU's total CO₂ emissions, with road traffic making up 72% of that total.

Rising concerns about climate change, and the desire to reduce the CO₂ and other emissions that are causing it, has led the EU to introduce stricter emission targets for new cars and vans and approve a proposal to cut CO₂ emissions from new trucks by 30% by 2030 compared to 2019 levels. This means the need to switch to electric vehicles has taken on growing urgency.

When considering the cradle-to-grave lifecycle of a vehicle (including production and end-of-life disposal), EVs are already often cleaner than those powered by fossil-based fuels – and the balance will be tipped even further in their favor as the availability of clean, renewable electricity continues to increase.

For now, squeezing maximum efficiency from EVs, and avoiding the use of expensive specialized materials to build the motors that power them, is a key enabler for the switch to large-scale adoption.

III. Direct versus indirect cooling

In electric motors, the current flows in the windings, which are made of copper because of its low resistance. Electrical losses are an unavoidable obstacle and increase with the amount of current; for example, doubling the current quadruples the losses, which increases heat in the motor. In addition, magnetic losses heat both the stator and the rotor, further increasing the heat load. This heat can be extracted by direct or indirect cooling.

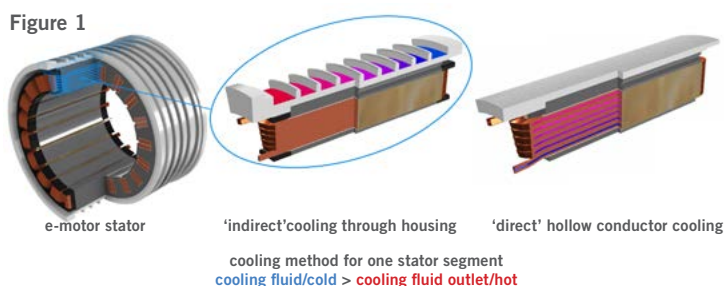


Figure 1. Exemplary stator segment to visualize the general concept of state-of-art indirect cooling and the innovation of direct cooling through hollow conductors. Source: hyperdrives GmbH.

Most smaller electric motors today use indirect cooling, as they have for more than a century. A cooling fluid, water glycol, in the motor housing transports heat to a heat exchanger where it is re-cooled by air before being returned to the housing. The drawback of this method is its limited heat flow, especially for the wire enamel and slot liner, which have low thermal conductivity. This can be seen in Figure 1, taken from a thermal simulation of conventional indirect liquid cooling.

Figure 2

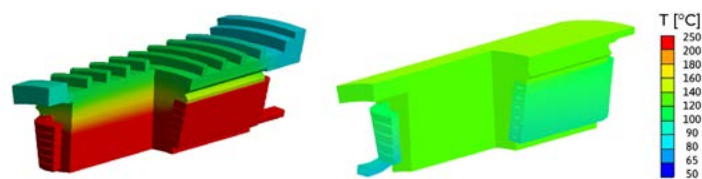


Figure 2. Thermal simulation image showing the efficient cooling enabled by hollow conductors in an exemplary stator segment. Source: hyperdrives GmbH

As can be seen in Figure 2, taken from a thermal simulation of direct liquid cooling, direct stator cooling does not suffer from this problem. Using hollow conductors enables more efficient cooling as a cooling fluid or gas can be circulated in direct contact with the inside of the conductor. This allows for significantly higher stator currents and therefore smaller, more compact motors.

The big benefit of this method is that it treats the heat where it is generated, in the winding itself instead of in the housing. Effective cooling of the stator also ensures a lower rotor temperature, enabling the use of less expensive lower temperature graded permanent magnets.

Hollow conductors in electric motors are not a fundamentally new invention; they have been used for decades in large electric motors and generators, where the distance between the heat-generating winding and the coolant in the housing is too long for indirect cooling to be effective. However, with smaller motors the challenge up to now has been the dimensions of the conductor, as little as 1 mm, which increases manufacturing complexity and negatively affects coolant circulation causing high pressure demand on the pump side. Solving this challenge has paved the way for the use of hollow conductors in smaller direct-cooled motors.

The challenges of direct cooling

Direct cooling is not without its challenges, however. These include the high pressure drop in the hydraulic cooling system, the need to seal the connection joints and to use a cooling fluid with sufficient insulation properties, and more complex manufacturing processes. With hollow conductors there is also a limit to how small the motor can be made. Frequency is also limited as increased frequency would require even finer conductors. However, direct-cooled motors typically allow for frequencies of up to 1 kHz, which is suitable for consumer EVs and commercial vehicles.

IV. The benefits of direct-cooled electrical motors

Direct cooling offers the highest power density for an electric motor – especially continuous power output – alongside excellent efficiency over a broad range. Partial load is the most realistic use case for an electric motor, and tests conducted in an evaluation prototype created by German company hyperdrives GmbH have shown that a direct-cooled motor offers approximately twice the power output at peak load and three times the power output at continuous steady-state operation compared to a standard, indirect-cooled motor.

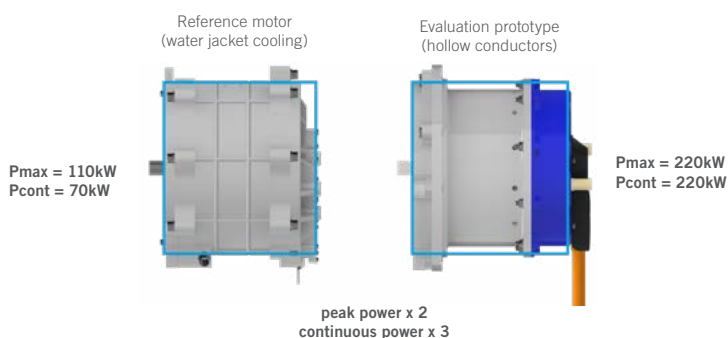


Figure 3. Comparison of peak and continuous power between an indirect and a direct-cooled electric motor. Source: hyperdrives GmbH

The higher efficiency of direct-cooled motors means that smaller power units can be used in EVs, saving both weight and costs. It also enables the use of electric powertrains in larger vehicles such as trucks. Direct-cooled motors can use standard, cost-effective materials, including magnets, providing higher performance at a lower cost – a key enabler for the wider adoption of EVs.

The benefits of direct-cooled motors in various applications (100 kW–2 MW)

Automotive

- Excellent efficiency at partial loads
- Outstanding performance with standard materials
- Competitive performance-to-cost ratio

Commercial vehicles

- Excellent robustness and durability due to reduced stator temperature level, especially for the winding insulation
- High efficiency across the full operation range and high constant power density
- Machine size optimized for minimal total cost of ownership

Aviation

- Unparalleled overload capability optimized for constant flight operation and maximum system efficiency
- Low noise and vibration

Industrial applications

- High torque and power density and excellent efficiency across the full operation range
- Tight control of motor temperature
- Low torque ripple, noise, and vibration

V. Luvata and hollow conductors

Luvata has been manufacturing hollow copper conductors for demanding applications for nearly 60 years and provides automotive-grade material for high-performance products. Luvata can manufacture hollow conductors of various sizes, with short lead times. Sizes start from 3 x 2 mm (< 0.07 kg/m) and continue up to above 100 mm in width (> 30 kg/m). Luvata also supplies conductors in small quantities for testing and prototyping.

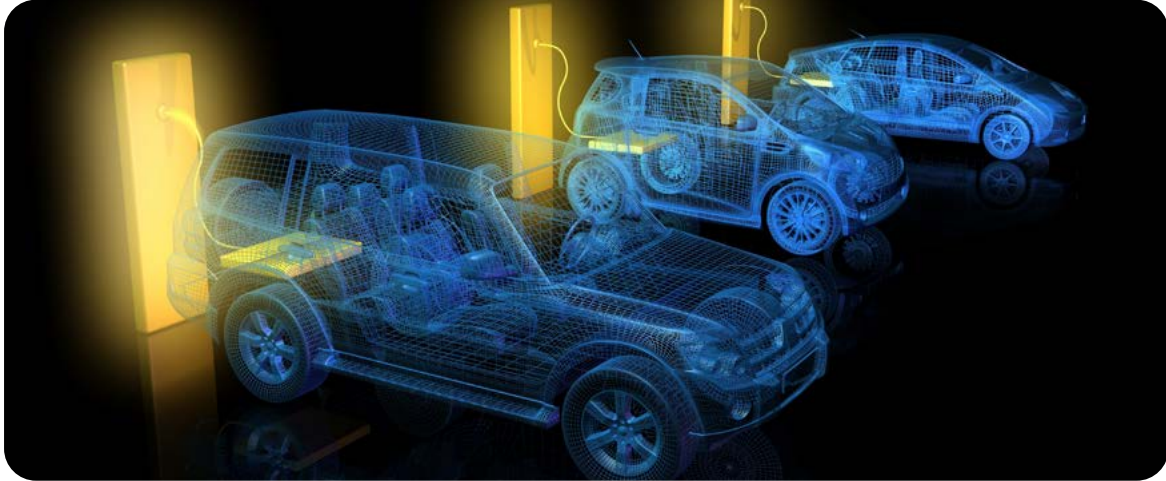
Luvata conductors are made of high purity oxygen-free copper and are ideal in applications where high electrical and thermal conductivity are essential requirements. Oxygen-free copper can be joined with all welding and brazing methods and is suitable for manufacturing processes requiring extreme deformability. Oxygen-free copper also has better bending properties compared to the standard ETP grade often used in stator windings.

Hollow conductors have many applications, including:

- Electric vehicles (EVs) / stator direct cooling
- MRI devices / gradient coils
- Cancer treatment: proton therapy and boron neutron capture therapy
- Electric magnets for high-energy physics applications
- Particle accelerators
- Generators
- Induction furnaces / induction heating and melting
- Plasma research devices
- Electrodynamics vibration test systems
- Ion implantation units for the microcircuit industry
- High-gradient separators
- Waveguides
- Liquid-cooled transformers

VI. Conclusion

Direct-cooled electric motors using hollow conductors offer many advantages over standard electric motor designs. A direct-cooled electric motor enables higher power density and can be constructed from standard materials in smaller sizes. All this will be needed to scale-up the use of EVs and make them as affordable, efficient, and sustainable as possible – **and Luvata has the experience, expertise, and solutions to help make it happen.**



About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services, serving industries such as renewable energy, automotive, healthcare, and power generation and distribution. The company's continued success is attributed to its longevity, technological excellence, and strategy of building partnerships beyond metals. Luvata employs approximately 1,400 staff in six countries, and its partners include ABB, CERN, Siemens, and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

About hyperdrives

hyperdrives is a Munich-based start-up that designs and tests electric drivetrains using oxygen-free copper hollow conductors from Luvata.

Get in contact with the hyperdrives team through **info@hyperdrives.de** or visit **www.hyperdrives.de** to find out more.

Find out more at luvata.com/products/hollow-conductors

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(1) A. Reinap, M. Gabassi, M. Alaküla, M. Andersson, "Assessment of cooling integration with direct cooled windings", IEEE International Conference on Electrical Systems for Aircraft, Railway, Ship Propulsion and Road Vehicles & International Transportation Electrification Conference, 2018

Data Sheet

GENERAL DESCRIPTION
– SUBJECT TO CHANGES OR DEVIATIONS

Oxygen-free Electronic Grade Copper Cu-OFE – Luvata Alloy OFE-OK®

Alloy description

Luvata OFE-OK® oxygen-free copper is min 99,995% pure copper for most demanding applications. It is used in special applications where risk of evaporation of low melting elements is not allowed and extreme material integrity and homogeneity are required. It can be joined with all welding and brazing methods and it is suitable for manufacturing processes requiring extreme material deformability. Luvata OFE-OK can be supplied with guaranteed Residual Resistance Ratio (RRR) value up to min 400.

Typical applications:

- Vacuum components
- Laser mirrors
- Cryogenic applications
- Other applications where ultra-high purity is needed

Products / shapes:

Profile tubes, round tubes, round rods and wire. Corresponding EN- and ASTM norms for different products are as follows:

- EN13604 – Copper and copper alloys.
Products of high conductivity copper for electronic tubes, semiconductor devices and vacuum applications.
- ASTM F68 – Standard specification for Oxygen-Free Copper in Wrought Forms for Electron Devices.

Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition * %	EN – CEN/TS 13388:2008	ASTM / USA	GOST / Russia ГОСТ 859:2001
OFE-OK	Cu 99,995 %	Cu-OFE / CW009A	CDA C10100	M00B (M00Б)

* Other elements as EN 13604 and ASTM B170 max %: Ag 0.0025, As 0.0005, Bi 0.0001, Cd 0.0001, Fe 0.001, Mn 0.00005, Ni 0.001, O 0.0005, P 0.0003, Pb 0.0005, S 0.0015, Sb 0.0004, Se 0.0002, Sn 0.0002, Te 0.0002, Zn 0.0001

Physical properties:

Density kg/dm ³	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,94	0,0000177	385	1083

Mechanical properties – typical values:

	Soft temper	Half-hard temper	Hard temper
Hardness HV	35 – 65 HV	70 – 95 HV	85 – 115 HV
Tensile strength	200 – 220 N/mm ²	250 – 350 N/mm ²	260 – 400 N/mm ²
0,2% yield strength	35 – 65 N/mm ²	180 – 280 N/mm ²	220 – 380 N/mm ²
Elongation	min. 40 %	min. 12 %	min. 5 %

Electrical and thermal properties – typical values:

Electrical conductivity	vol	% IACS *	min 102,1
	mass	%IACS	min 101,5
	MS/m		min 59,2
Electrical resistivity	vol	Ω mm ² /m	max 0,0169
	mass	Ω g/m ²	max 0,1510
Thermal conductivity (20 °C)	W / Km		391

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Excellent	Excellent	Good	Good	Excellent

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Data Sheet

GENERAL DESCRIPTION
– SUBJECT TO CHANGES OR DEVIATIONS

Oxygen-free Silver Bearing Copper – Luvata Alloy HK003, HK015

Alloy description

Luvata HK silver bearing oxygen-free copper is a high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are essential. Small amount of alloyed silver increase the softening temperature of pure copper and the mechanical properties at elevated temperature are higher than in pure copper. It can be joined with all welding and brazing methods and it is suitable for manufacturing processes requiring extreme deformability.

Typical applications:

- Commutators
- Generator rotor and stator windings
- Applications where creep properties and higher softening temperature than in OF-OK® are needed
- Other applications where high electrical and thermal conductivity are needed

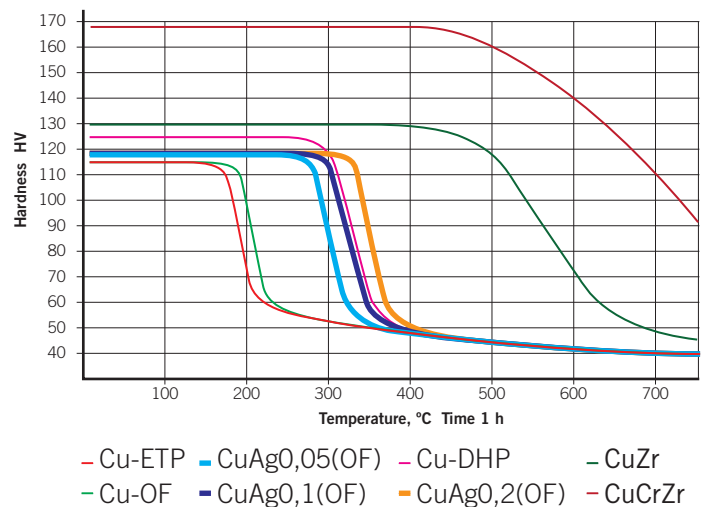
Products / shapes:

Profile tubes, round tubes, round rods, wire and strip coils, rectangular bars and solid profiles. Corresponding EN-norms for different products are as follows:

- EN 13600 – Copper and copper alloys.
Seamless copper tubes for electrical purposes.
- EN 13601 – Copper and copper alloys.
Copper rod, bar and wire for general electrical purposes.
- EN 13605 – Copper and copper alloys.
Copper profiles and profiled wire for electrical purposes.

Softening behaviour – resistance against softening:

Room temperature hardness is presented in the following figure as a function of annealing temperature. Material at hard or aged temper.



Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition* Ag –content %	EN – CEN/TS 13388:2008	ASTM / USA
HK003	0,03 – 0,05 %	CuAg0, 04 (OF) / CW017A	CDA C104 OFS
HK015	0,085 – 0,12 %	CuAg0,10 (OF) / CW019A	CDA C107 OFS

* Other elements max %: Bi 0,0005, others total 0,0065

Physical properties:

Density kg/dm ³	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,94	0,0000177	385	1083

Mechanical properties – typical values:

	Soft temper	Half-hard temper	Hard temper
Hardness HV	35 – 65 HV	70 – 95 HV	85 – 115 HV
Tensile strength	200 – 220 N/mm ²	250 – 350 N/mm ²	260 – 400 N/mm ²
0,2% yield strength	35 – 65 N/mm ²	180 – 280 N/mm ²	220 – 380 N/mm ²
Elongation	min. 40 %	min. 12 %	min. 5 %

Electrical and thermal properties – typical values:

			HK003	HK015
Electrical conductivity	vol	% IACS *	min 99,5	min 99,5
	mass	%IACS	min 98,9	min 98,9
	MS/m		min 57,7	min 57,7
Electrical resistivity	vol	Ω mm ² /m	max 0,0173	max 0,0173
	mass	Ω g/m ²	max 0,1549	max 0,1549
Thermal conductivity (20 °C)	W / Km		388	388

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

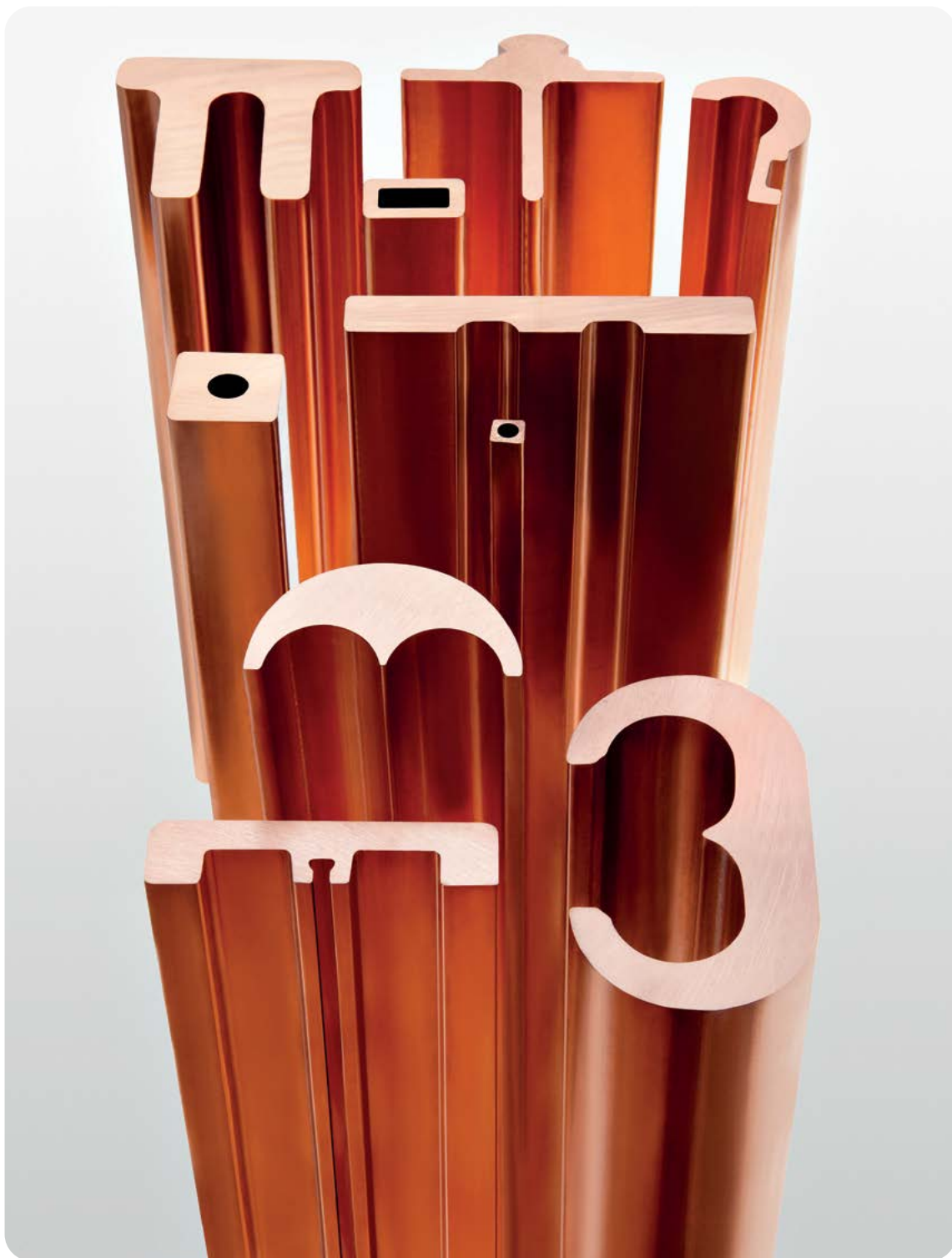
Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Excellent	Excellent	Good	Good	Good

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Oxygen-free Copper

Luvata's highly malleable and consistent oxygen free copper offers superior electrical and thermal conductivity.

LUVATA
Partnerships with a Promise



The Purest Commercial Copper Grade

The purity of oxygen-free copper makes it exceptionally conductive. This quality, along with its malleability and metallurgical consistency, makes it ideal for the most precise industrial and scientific applications. It can be found in superconducting wires, circuit boards, vacuum interrupters, X-ray tubes, laser mirrors and scientific instruments, where precision is critical, and the copper needs to be of a consistently high quality.

Purity is only part of the story

It is the purity of oxygen-free copper that makes it superior to other commercial copper alloys, but the people behind the product are equally important. Whether it is a case of needing high electrical conductivity, a material that resists deformation in the manufacturing process, or secure joining by brazing, our people have the experience and insight to know when and how oxygen-free copper can be used to its best advantage. We believe in forging strong, long-term relationships and engaging fully with our customers' businesses, helping them solve problems, innovate and design tailor-made solutions

Our in-house casting allows us to fully control the production process and the quality of our end products. We have just invested in a state-of-the-art hot extrusion line in Finland including the latest in material handling robotics and automation, allowing for even greater product customization.

"As we all strive towards a greener, more digitized future, oxygen-free copper will play an increasingly important role. It is used in scientific applications such as fusion research for renewable energy, and can be found in the superconductors at the CERN particle accelerator. It is also central to the green transition, through the production of renewable electricity and the development of e-mobility solutions. In fact, we see unlimited potential for the future of oxygen-free copper in building a more sustainable future, including solving challenges that haven't even been thought of yet."

We are serious about sustainability. Every strategic decision we make takes into account our environmental impact, and we have pledged to become carbon neutral by 2045. We believe oxygen-free copper as fundamental to our vision of a future which benefits people, society and the Earth."

President & CEO
Pekka Kleemola

Luvata OF-OK® Copper

Luvata Oxygen-Free Copper was specifically developed for electrical, electronic and heat transfer applications. It is an extremely pure copper alloy, with a maximum oxygen content of 5 ppm (0.0005%) and has a unique combination of high-performance properties.

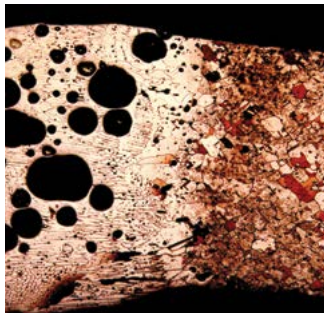
Consistency

High quality raw materials, melting and casting in electric furnaces, and a strictly controlled manufacturing process, guarantee the material is consistent from batch to batch.

Consistency means less time consumed for machine set-up and less waste. By using silver bearing oxygen-free copper (alloying copper with a small amount of silver) its softening temperature can be increased significantly without compromising its high conductivity or formability. Silver bearing Luvata OF-OK® (Ag) is perfectly suited to commutators, generator rotor bars and welding components.

Electrical and Thermal conductivity

As a result of its purity, Luvata OF-OK® copper has high electrical conductivity (typically 100 – 102% IACS) and high thermal conductivity (390 W/Km). The higher conductivity values compared with other materials mean that equipment made of Luvata OF-OK® copper operates more efficiently and at lower temperatures, which increases their service life and reliability.



Hydrogen embrittlement in TIG welded, ETP (left) versus Luvata's TIG welding seam in oxygen-free OF-OK® copper (right)



Immunity to Hydrogen embrittlement

At high temperature, hydrogen ions can diffuse into copper and form water with the oxygen atoms in the copper. The resulting water vapor has a high enough pressure to form pores in the copper known as hydrogen embrittlement. It can occur when brazing or welding copper components in a moist environment or within a reducing atmosphere containing hydrogen. In commercial copper grades which usually contain 200-600 ppm (0.02-0.06%) of oxygen, the result is a completely brittle metal. In Luvata OF-OK® copper, we guarantee a maximum of 5 ppm (0.0005%) oxygen content, which makes it immune to hydrogen embrittlement and ensures safe joining with all brazing and welding methods.

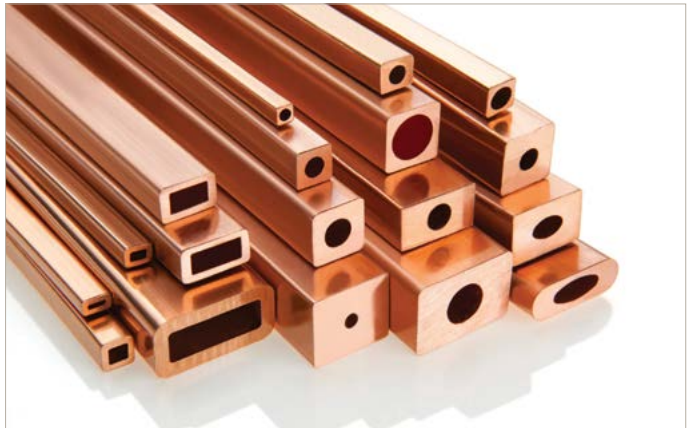
Applications

Luvata OF-OK® is the ideal material for a large number of applications:

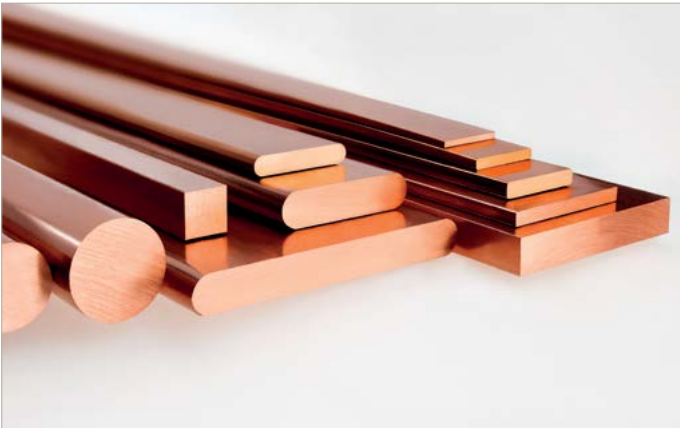
- Rotor bars and stator winding for power generators
- Components for electrical equipment
- Coils for electrical magnets and induction furnaces
- Connectors
- Heat sinks for semiconductor bases



Source: XFEL
XFEL corrector quadrupole



Luvata hollow conductors, OF-OK® and OFE-OK®



Luvata rods, bars and busbars, OF-OK® and OFE-OK®

Luvata OFE-OK® Certified Grade Copper

Luvata's OFE-OK® oxygen-free electronic grade copper is tested at every stage of the manufacturing process, from selecting the raw material to signing the inspection certification. That is why we call it Certified Grade Copper.

A higher grade material for specialist applications

For the most demanding electronic, cryogenic and optical applications, a higher grade oxygen-free copper is needed. Luvata OFE-OK® copper is a high-purity copper grade with total impurities of less than 40 ppm (0.0040%), with no single impurity exceeding 25 ppm (0.0025%). OFE-OK® copper is manufactured, controlled and inspected separately from standard Luvata OF-OK® copper. For the certified grade, only specially selected cathode copper is used and every ingot is inspected to ensure compliance with ASTM F68 Oxygen-Free Copper in Wrought Forms for Electron Devices specification and other international specifications for OFE copper.

Vacuum Applications

Luvata OFE-OK® Certified Grade Copper has an extremely low content of highly volatile low melting elements such as Phosphorus (P), Lead (Pb), Sulphur (S), Zinc (Zn), Cadmium (Cd) and Mercury (Hg). In a number of vacuum applications in the electronic industry, glass is directly bonded to copper, requiring an adherent oxide film on the copper surface. This oxide film can be rendered non-adherent and brittle if phosphorus, which is often used as oxidant, is present. To prevent this, Luvata OFE-OK® contains less than 3 ppm (0.0003%) of phosphorus, allowing for a direct bond to glass and ceramic materials. The low volatility and limited phosphorus content in Luvata OFE-OK® ensure the smooth functioning and long service life for vacuum components such as vacuum interrupters, X-ray tubes and radio transmitters.

Integrity

Luvata OFE-OK® is tested during all steps of the manufacturing process; casting, extrusion, drawing and at the end of the process to ensure the integrity of the material. Even small impurities or defects from extrusion may cause scattering of the beam from the optical component and cause rejection after a great deal of time and effort has been spent. Using OFE-OK® ensures the highest possible yield in the process. All Luvata OFE-OK® copper carries a test certificate guaranteeing the impurities are below the specified limits, electrical conductivity is at least 101.5% IACS and the material integrity complies with the requirements of the most demanding applications.

Cryogenic Properties

At cryogenic temperatures, the electrical and thermal properties of Luvata OFE-OK® copper are superior to any other copper grade. With the high purity of Luvata OFE-OK® combined with the special thermal treatment in the manufacturing process, the high value RRR (Residual Resistance Ratio) of Luvata OFE-OK® is guaranteed. RRR values up to 2000 are available with specific high purity copper grades.

Since RRR is very sensitive for common processing treatments, Luvata experts are pleased to advise the best thermomechanical treatment to meet the optimum RRR value in the final component. Unlike oxygen-containing copper alloys with high RRR values, Luvata OFE-OK® can readily be electron beam welded or annealed in a reducing atmosphere without the risk of hydrogen embrittlement.

On request, Luvata also provides copper materials having RRR within a specific range and alloyed copper grades with very low RRR value.



Source: Kugler GmbH
Laser mirrors



Source: Lohmann X-Ray GmbH
X-ray tube

Technical data

Alloys

Alloy	Luvata Alloy	Purity %	EN	ASTM
Cu-OF	OF-OK®	99.95 min	CW008A	C10200 OF
Cu-OFE	OFE-OK®	99.99 min	CW009A	C10100 OFE, ASTM F68
CuAg0.04(OF)	HK003	Cu + Ag min 99.95	CW017A	C10400 OFS
CuAg0.10(OF)	HK015	Cu + Ag min 99.95	CW019A	C10700 OFS

Size range

Rods mm	Mill Length mm	Temper
3 – 25	3500 – 4000	hard, soft annealed
25 – 60	3500 – 4000	hard, soft annealed
50 – 125	2000 – 3000	hard, soft annealed
125 – 190	1000 – 2000	extruded, forged
190 – 850	500 – 1500	cast, forged

Busbars

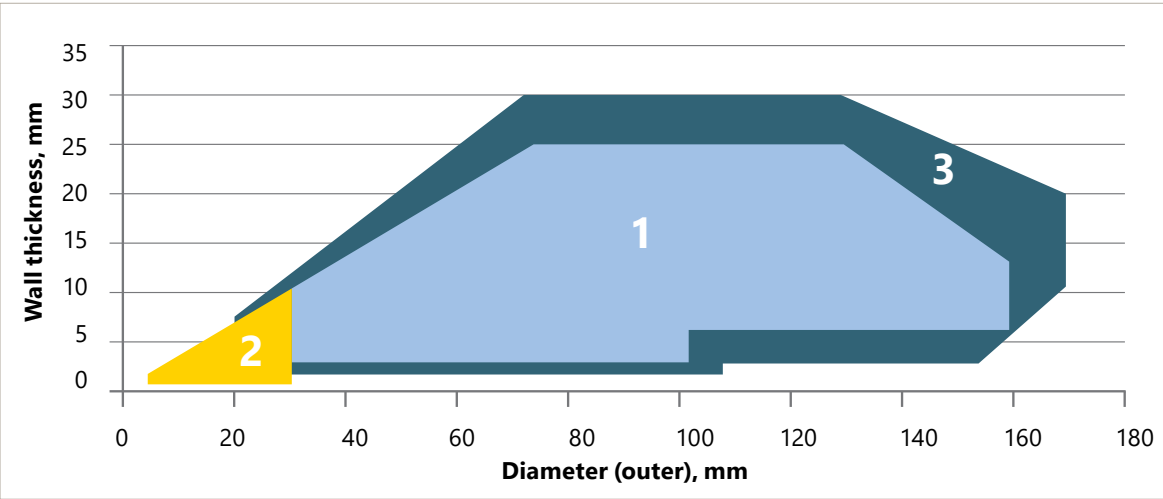
Width mm	Min Thickness mm
10 – 20	2
20 – 40	3
40 – 70	4
70 – 250	5

Tubes

- Deliver forms:**
- Straight lengths up to 12 m
 - Loose coils up to 120 kg
 - Mileon® continuous coils in wooden reels up to 2500 kg
 - LWC (Level Wound Coil) coils
 - Forged tubes up to OD 800 mm

Hollow conductors

- Size range:** Different shapes from 3 x 2 mm up to 100 x 100 mm.
- Delivery forms:**
- Straight lengths up to 16 m
 - Loose coils
 - Mileon® continuous coils in wooden reels up to 2500 kg
 - Pancake coils
 - LWC (Level Wound Coil) coils



Region No.1 – tubes in straight length.
Region No.2 – tubes in coil.
Region No.3 – special dimensions (available upon request). Other dimensions – check availability from Luvata.

Luvata - Partnerships with a Promise

Working for People, Society and the Earth

We all want to ensure a sustainable future for our planet. As a manufacturer, we know that we have both an opportunity and a responsibility to contribute positively to the changes needed to secure that future. We have stated that our vision is to be the leading provider of copper solutions that benefit people, society and the earth, and this is a serious commitment. We are active in markets such as e-mobility, healthcare, electrification and the green transition, all of which have a need for copper products. We believe that our innovations will help drive these markets forwards, making green solutions more easily accessible to all.

We also recognize the need for our operations to be sustainable. To this end, we are striving to reduce greenhouse gas emissions, aiming to be using 100% renewable electricity by 2035 and to be fully carbon neutral by 2045.

Helping our customers grow and thrive

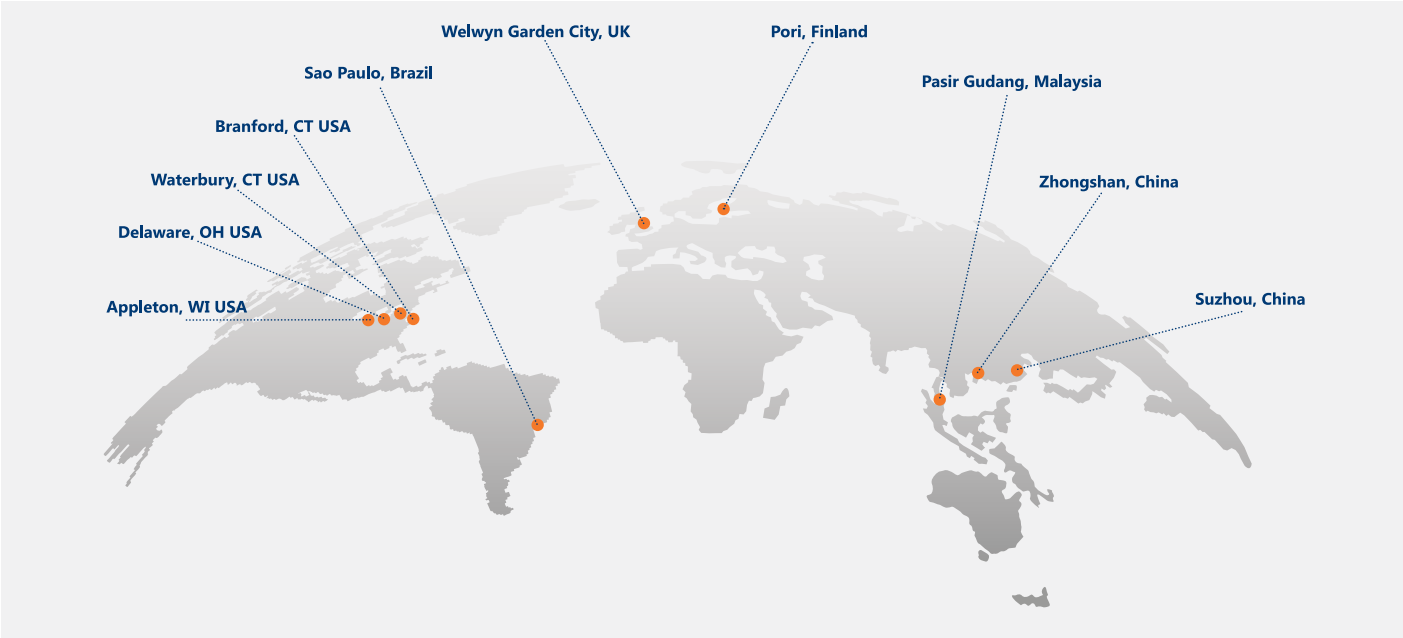
We have always worked in partnership with our customers, getting to know their businesses and collaborating on custom-made solutions. We take great satisfaction in seeing our customers thrive, and knowing that we played a part in their success.

Where we believe we can make a real difference is in sharing our robust manufacturing capabilities, technology and material knowledge, and by using our unique expertise to develop technically-demanding products. Together with an emphasis on lean production, innovation and value-added solutions, we work hard to ensure that we are the partner of choice for our customers, every time.

Partnerships with a Promise

At Luvata, we deliver on our promises. We have pledged always to treat people with respect, keep an open mindset and work together with others.

Our global footprint stretches across four continents. Our diversity of locations, cultures and markets gives us access to a wealth of knowledge and expertise that continues to grow. It also means that we are local to our customers, wherever they are, so we can continue to forge the strong relationships we are known for.



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About Luvata

Luvata is a world leader in copper manufacturing and related engineering services. It brings together people, innovation and technology to create solutions in areas such as e-mobility, healthcare, and in electrification & the green transition. The company's success is driven by innovation, lean production and value-added sales, and its vision is to provide copper solutions which benefit people, society and the earth. Employing over 1,400 staff on four continents, Luvata works in partnership with customers such as ABB, CERN and Siemens. Luvata is a group company of Mitsubishi Materials Corporation.



LUVATA
Partnerships with a Promise



Hollow Conductors

Copper hollow conductors for the most demanding applications, from MRI scanners to electric vehicles and power generators.

LUVATA
Partnerships with a Promise



A history of delivery and innovation

At Luvata, we have been producing copper hollow conductors for over half a century. In that time, we have built a track record for reliability, adaptability and constant innovation. Our hollow conductors can be found in MRI scanners, particle accelerators, semiconductors, nuclear research facilities and the stator winding of 1600 MW power generators. Our innovations include Mileon®, the longest jointless hollow conductor available anywhere. Developed to meet the demand for seamless magnetic windings, it is uniquely durable and reliable. We also have in-house billet casting, giving us complete control over the quality of our products.

Our strength lies in our relationships

We believe in forging strong, long-term relationships with our customers, our suppliers and our staff. Our manufacturing staff are fully engaged with our customers' businesses, and have an in-depth understanding of both the end applications and the technical requirements. This enables us to work closely with our customers to problem-solve, innovate and design customized solutions to specific projects. Our employees typically stay with us for many years, ensuring consistency and quality that is second-to-none. All of which enables us to add value over and above a traditional supplier relationship, and build partnerships which really last.

"As a global company, we have the power to contribute positively to a better future, and that is a responsibility we take seriously. The megatrends in our markets mean the use of copper products can only increase, and our hollow conductors have an important role to play in the decarbonization of society. We all want to see a more sustainable future, and Luvata is determined to be a force for good in the development of e-mobility, the green transition, electrification and in healthcare and scientific research."

The world is changing rapidly, and with our history of innovation and collaboration, we are well placed to develop technically-demanding products which deliver exactly what is needed and make the most of the resources available to us. This is what drives us, and this is why we have pledged Partnerships with a Promise. We are determined to benefit people, society and the Earth and play our part in building a better tomorrow."

President & CEO
Pekka Kleemola

Bespoke hollow conductors to meet every need

We have been manufacturing copper hollow conductors for over sixty years, helping to build the most demanding scientific applications, MRI scanners and power generators. Much of what we do is tailor-made, and delivered to the most exacting requirements.

The specifications for hollow conductors change all the time. Although most applications call for square hollow conductors with round or rectangular holes in standard sizes, there is a growing need for alternative outer dimensions (OD) or hole sizes. At Luvata, we have been asked for many different shapes and sizes, and have always delivered the perfect solution.

High energy physics (HEP) magnets are getting bigger, and need larger hollow conductor ODs and hole sizes. Meanwhile, for the healthcare industry, where turns and teslas are everything, smaller ODs and hole sizes are required. And we are often asked to produce non-standard cross-sections in a variety of shapes and sizes. With the widest range of tools on the

market, we have the flexibility to adapt to our customers' needs.

Looking to the future, we are applying our expertise to the growing demand for Electric Vehicles (EVs). Direct-cooled motors with hollow conductor stators have several advantages over the indirect-cooled motors currently being used. Our solution enables higher power density and can be constructed from standard materials in smaller sizes, making EVs even more efficient and more affordable.

We help our customers build particle accelerators, plasma research devices, MRI scanners and ion implantation units that are efficient and cost-effective.

Some of the end uses for our products

- Electric vehicles (EVs) / Stator direct cooling
- Particle accelerators
- MRI scanners
- Semiconductors
- Ion implantation units for the microcircuit industry
- Plasma research devices / fusion energy
- Heating / cooling in battery packs
- Generator windings
- Electrodynamical vibration test systems
- Induction furnaces
- Electromagnetic stirrers
- High gradient separators



Wind farm generator windings



Particle accelerator



Electric vehicle



MRI scanner

Extremely long coils help eradicate leakage, and mean reductions in waste and packaging.

Every joint in a water-cooled application is a potential cause of leakage. So the longer each individual coil is, the more the risk is reduced. We developed our extremely long Mileon® coils with the aim of eliminating the joints altogether. Now, Mileon® coils, rather than being measured in meters, are measured in hundreds, or even thousands of meters.

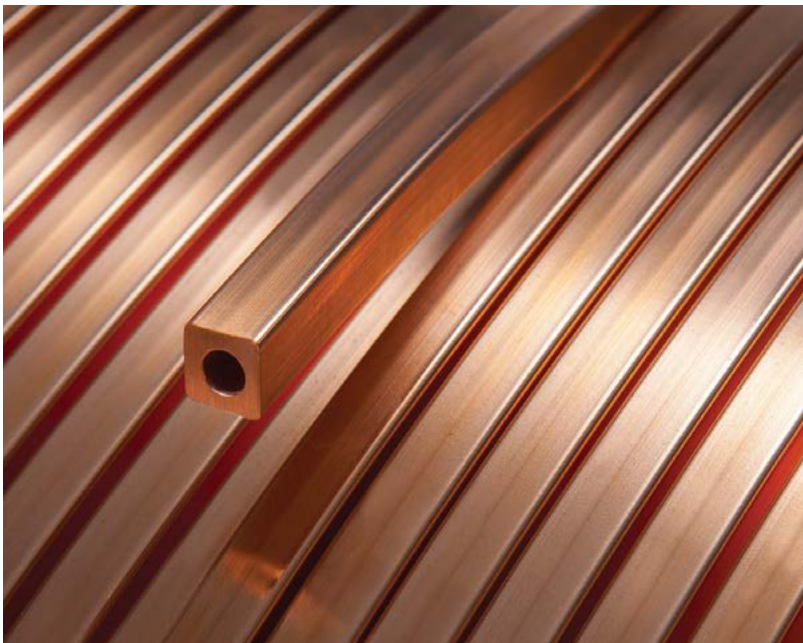
Mileon® coils are manufactured using only oxygen-free copper, and are subject to the same stringent quality criteria that we apply to our other hollow conductor products.

With Mileon® coils, less scrap is generated in the manufacturing process, and expensive copper is saved. The longer coils also provide increased flexibility in production compared with shorter coils that are cut to predetermined lengths. Mileon® coils are typically delivered on large wooden

spools, which require less packaging and are recyclable, and alternative environmentally friendly packaging options are also available.

The advantages of Mileon® coils

- No joints
- Highest quality copper
- Less copper wastage
- Precise delivery quantities



Technical data

All impurities reduce the electrical conductivity of copper, which is why we make Luvata hollow conductors from OF-OK® high purity oxygen-free copper. As a result of its purity, Luvata OF-OK® has typical electrical conductivity of 100-102 % IACS and thermal conductivity of 390 W/Km.

The maximum oxygen content of 5 ppm (0.0005%) makes the material immune to hydrogen embrittlement and ensures safe joining with all brazing and welding methods.

Alloys

Alloy	EN	ASTM	Electrical conductivity		Thermal conductivity
			MS/m	IACS%	W/Km
OFE-OK®	CW009A	C10100	58,6	101,5	391
OF-OK®	CW008A	C10200	58,3	100	390
CuAg.0.04(OF)	CW017A	C10400	57.7	99.5	388
CuAg0.10(OF)	CW019A	C10700	57.7	99.5	388

Hundreds of different conductors

Our long history in making hollow conductors has given us thousands of tools for creating all sizes of conductors within very short time scales. We can also supply conductors in small quantities for testing and prototyping purposes.

Hollow conductors

Size range: Different shapes from 3 x 2 mm up to 100 x 100 mm.

Delivery forms

Mileon jointless long lengths	up to 2500 kg depending on dimension, on a wooden reel
Loose coils, pancake coils or LWC coils	max 50 – 400 kg depending on dimension
Spools for small sizes	plywood or plastic spools
Straight lengths	up to 16 m



Scan QR code for access to our online tool list.

The OF-OK® used in our hollow conductors is cast in our own foundry in electric furnaces using selected high purity raw materials. This means we can control the whole manufacturing process from casting to signing the inspection certification, and ensure the consistency of our product from batch to batch.

When necessary, OF-OK® is alloyed with silver (Ag) to increase both the softening temperature and creep resistance of the alloy without any significant sacrifice of its other properties.

Please visit www.luvata.com/hollow-conductors, for an up-to-date tool list. If you don't find what you are looking for, contact us and our engineers will happily create a customized solution for you.

Just some of our hollow conductor profiles



Luvata - Partnerships with a Promise

Working for People, Society and the Earth

We all want to ensure a sustainable future for our planet. As a manufacturer, we know that we have both an opportunity and a responsibility to contribute positively to the changes needed to secure that future. We have stated that our vision is to be the leading provider of copper solutions that benefit people, society and the earth, and this is a serious commitment. We are active in markets such as e-mobility, healthcare, electrification and the green transition, all of which have a need for copper products. We believe that our innovations will help drive these markets forwards, making green solutions more easily accessible to all.

We also recognize the need for our operations to be sustainable. To this end, we are striving to reduce greenhouse gas emissions, aiming to be using 100% renewable electricity by 2035 and to be fully carbon neutral by 2045.

Helping our customers grow and thrive

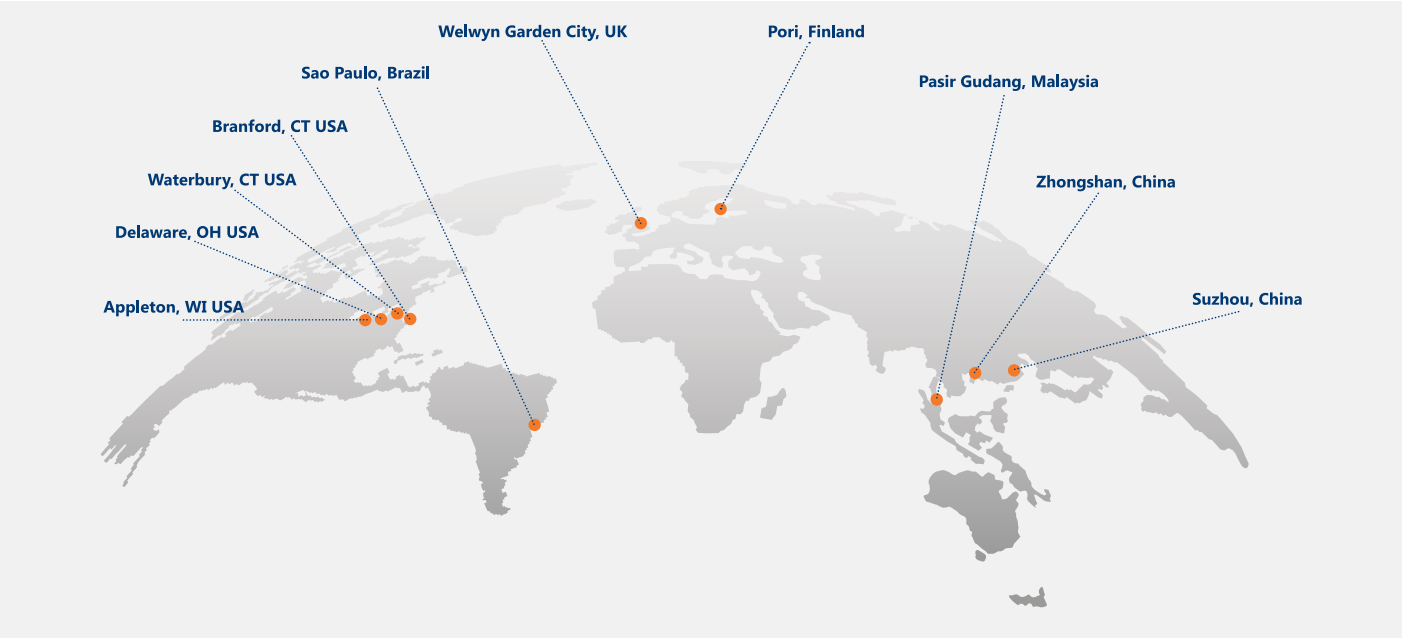
We have always worked in partnership with our customers, getting to know their businesses and collaborating on custom-made solutions. We take great satisfaction in seeing our customers thrive, and knowing that we played a part in their success.

Where we believe we can make a real difference is in sharing our robust manufacturing capabilities, technology and material knowledge, and by using our unique expertise to develop technically-demanding products. Together with an emphasis on lean production, innovation and value-added solutions, we work hard to ensure that we are the partner of choice for our customers, every time.

Partnerships with a Promise

At Luvata, we deliver on our promises. We have pledged always to treat people with respect, keep an open mindset and work together with others.

Our global footprint stretches across four continents. Our diversity of locations, cultures and markets gives us access to a wealth of knowledge and expertise that continues to grow. It also means that we are local to our customers, wherever they are, so we can continue to forge the strong relationships we are known for.



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LUVATA
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Data Sheet

GENERAL DESCRIPTION
– SUBJECT TO CHANGES OR DEVIATIONS

Oxygen-free Copper Cu-OF – Luvata Alloy OF-OK®

Alloy description

Luvata OF-OK® oxygen-free copper is high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are the essential requirements. It can be joined with all welding and brazing methods and it is suitable for manufacturing processes requiring extreme deformability.

Typical applications:

- Magnet windings
- Semiconductor components
- Electric motors
- Wave guide tubes
- Induction furnaces
- Electrical components
- Switchgear applications
- Generator material in rotor and stator windings
- Other applications where high electrical and thermal conductivity is needed

Products / shapes:

Profile tubes, round tubes, round rods, wire and strip coils, rectangular bars and solid profiles. Corresponding EN-norms for different products are as follows:

- EN 13600 – Copper and copper alloys.
Seamless copper tubes for electrical purposes.
- EN 13601 – Copper and copper alloys.
Copper rod, bar and wire for general electrical purposes.
- EN 13605 – Copper and copper alloys.
Copper profiles and profiled wire for electrical purposes.

Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition * %	EN – CEN/TS 13388:2008	ASTM / USA
OF-OK	Cu + Ag min. 99,99 %	Cu-OF / CW008A	CDA C10200

* Other elements max %: O 0,0005, Bi 0,0005, Pb 0,005

Physical properties:

Density kg/dm ³	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,94	0,0000177	385	1083

Mechanical properties – typical values:

	Soft temper	Half-hard temper	Hard temper
Hardness HV	35 – 65 HV	70 – 95 HV	85 – 115 HV
Tensile strength	200 – 220 N/mm ²	250 – 350 N/mm ²	260 – 400 N/mm ²
0,2% yield strength	35 – 65 N/mm ²	180 – 280 N/mm ²	220 – 380 N/mm ²
Elongation	min. 40 %	min. 12 %	min. 5 %

Electrical and thermal properties – typical values:

Electrical conductivity	vol	% IACS *	min 100,6
	mass	%IACS	min 100,0
	MS/m		min 58,3
Electrical resistivity	vol	Ω mm ² /m	max 0,0171
	mass	Ω g/m ²	max 0,1532
Thermal conductivity (20 °C)	W / Km		390

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Excellent	Excellent	Good	Good	Good

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